

Quantum Mechanics III

Lecture 4

Open Systems and Lindblad Generators

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$\dot{\rho} = -i[H, \rho] + \sum_j \gamma_j (L_j \rho L_j^\dagger - \frac{1}{2} L_j^\dagger L_j \rho)$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$T_1^{-1} = \gamma_{\downarrow} + \gamma_{\uparrow}$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$T_2^{-1} = 1/(2T_1) + \gamma_\varphi$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 4; Wiseman, ch. 3; QDC coherence block